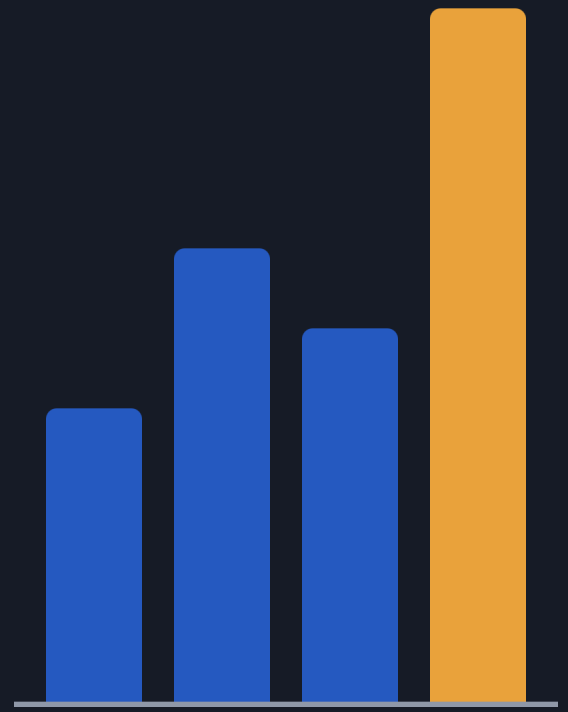
Power BI for Shiny users

From a messy client file to a dashboard that refreshes itself



The next 80 minutes

Clock	Part	You will see	Min
0:00	Opening	The demo company and its data	5
0:05	1 · Getting set up	Sign-up, Mac options, licences	10
0:15	2 · vs Shiny & Streamlit	One dashboard, three tools	15
0:30	3 · The workflow	Data → model → report → refresh	8
0:38	4 · Power Query & DAX	Clean a messy file, write measures	15
0:53	5 · Drill-down	Find why a region missed target	10
1:03	6 · How companies run it	Database to auto-refresh	12
1:15	Questions	Anything goes	5

Meet OutdoorCo, our demo company

$14.4M
revenue, Jan 2024 – Oct 2026

61,183
orders, in-store and online

12
stores across 5 regions

25
products in 5 categories

OutdoorCo_Sales.xlsx
clean star schema, 6 tables

raw_client_export_Q4_2025.csv
a deliberately messy file

client_drops/
monthly files plus a new November drop

code/
Shiny, Streamlit, DAX and M scripts

Synthetic data generated for this talk. Any resemblance to a real retailer is a coincidence.

01

Getting set up

Install, sign up, and what each licence actually buys you

Power BI is three apps, not one

Desktop

Free authoring app, **Windows only**. Most professionals build here and save **.pbix** files.

Service

The browser app at **app.powerbi.com**. Share, schedule refresh, and now build models and reports too.

Mobile

iOS and Android apps for viewing reports and getting alerts on the go.

On a Mac? Four ways in

Browser
Power Query, relationships and DAX now work on the web. Today's demos.

Parallels
Windows 11 on Apple silicon running Desktop

Cloud PC
Windows 365 or a virtual desktop from IT

Remote
Connect to a Windows machine that has Desktop

Signing up takes four steps

1

Open **app.powerbi.com** and choose **Start free**

2

Sign in with a **work or school** email

3

Land in **My workspace** with a Free licence

4

On Windows, add **Desktop** from the Microsoft Store

⚠ **The trap:** Gmail, Outlook.com and Hotmail addresses are rejected. Use your university or company email.

No work email? A Microsoft 365 trial gives you an @...onmicrosoft.com account that works. Blocked anyway? Your admin may have turned off self sign-up.

Free vs Pro vs Premium: the real difference is sharing

	Free	Pro	Premium Per User	Fabric capacity
Price	$0	$14 user/month	$24 user/month	Per capacity, from ≈$262/month (F2)
Build	My workspace only	Shared workspaces	Shared workspaces	Shared workspaces
Who can view	Only you	Other Pro users	Other PPU users	F64 and up: anyone with a Free licence
Scheduled refresh	8 per day	8 per day	48 per day	48 per day, more via API
Model size	1 GB	1 GB	100 GB	Depends on capacity size
Good for	Learning	Teams	Power users	Whole organization

“Publish to web” makes a report **public on the internet**. Never use it for real data.

02

One dashboard, three tools

Shiny and Streamlit write it; Power BI drags and drops it

The dashboard we will build three times

Filters

Year

2024 2025 2026

Region

Mountain
Pacific
Southwest
South
Midwest

Category

All 5

Revenue
$14.39M

Orders
61,183

Avg order value
$235.16

Gross margin
48.2%

Monthly revenue



Jan 2024 Oct 2026

By category

Camping $4.10M
Hiking $3.64M
Apparel $2.65M
Water $2.56M
Climbing $1.44M

By region

Mountain $3.92M
Pacific $3.83M
Southwest $2.89M
South $2.36M
Midwest $1.39M

Same dashboard, very different amounts of code

R SHINY

65

lines of R

```
f ← reactive({
  df ▷ filter(
    Year %in% input$year,
    Region %in% input$region)
})
output$rev ← renderText(
  dollar(sum(f()$Revenue)))
```

Plus UI layout, server logic and a host such as Posit Connect

STREAMLIT

44

lines of Python

```
f = df[df.Year.isin(years)
    & df.Region.isin(regions)]
revenue = f.Revenue.sum()
k1.metric("Revenue",
  f"${revenue:,.0f}")
```

Plus layout, caching and a server or Community Cloud

POWER BI

5

DAX measures, zero layout code

```
Revenue =
  SUM ( Sales[Revenue] )

Gross Margin % =
  DIVIDE ( [Revenue]
    - [Total Cost], [Revenue] )
```

Slicers, cross-filtering and drill-down are built in. Layout is drag and drop.

Non-blank, non-comment lines in shiny_app.R and streamlit_app.py from the talk pack

What each tool does for you

	R Shiny	Streamlit	Power BI
Interactivity	Whatever you code	Whatever you code	Cross-filter, drill, tooltips built in
Custom logic	All of R	All of Python	DAX, plus R or Python visuals
Data model	Joins in your code	Joins in your code	Relationships and measures
Sharing	Posit Connect, shinyapps.io	Community Cloud, own server	Service, Teams, mobile app
Refresh	Your own scheduler	Your own scheduler	Built in, 8 or 48 times a day
Security	You write it	You write it	Row-level security roles
Version control	Git, natively	Git, natively	PBIP project files and Git integration
Cost	Free, plus hosting	Free, plus hosting	Per user or per capacity

When to reach for which

Power BI when…

- Many business users need to view it
- Data already lives in Microsoft tools
- It must refresh on a schedule, securely
- It is mostly KPIs, trends and breakdowns

Shiny or Streamlit when…

- You run models or simulations live
- The interface is unusual or bespoke
- Users type inputs that change the logic
- You want open source and no licences

Most teams use both. R or Python does the modelling upstream and writes results to a table; Power BI is the reporting layer everyone opens.

03

The Power BI workflow

Seven steps from raw data to a report that refreshes itself

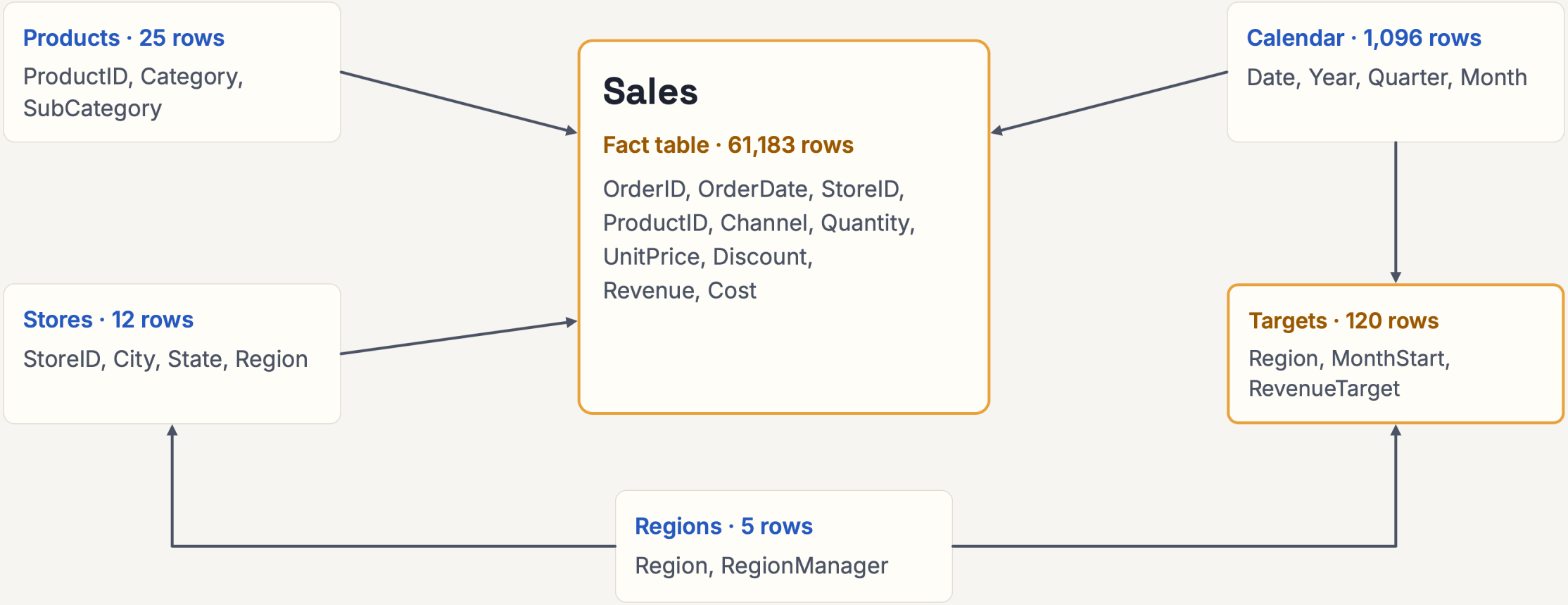
Seven steps, and you already know their R twins

1	2	3	4	5	6	7
Get data	**Transform**	**Model**	**Measure**	**Visualize**	**Publish**	**Refresh**
100+ connectors	Power Query	Relationships	DAX	Report canvas	Workspace	Schedule
`read_csv()`	`dplyr, tidyr`	`left_join()`	`summarise()`	`ggplot2, ui`	`Posit Connect`	`cron`

Steps 1–4 = the semantic model. Data plus business logic, built once.

Steps 5–7 = reports. Many reports can share one model.

A star schema sits behind the report



Arrows show filter direction: one row on the arrow's tail filters many rows at its head. Amber boxes are fact tables.

04

Power Query and DAX

Clean the mess first, then teach the model to calculate

The client file is a mess. Power Query cleans it.

3
junk lines above the header

10
blank rows

25
duplicate rows

10
spellings of 5 regions

3
date formats in one column

149
product names in CAPITALS

24
missing quantities

$
money stored as text, “$1,234.56”

1
column holding city and state

APPLIED STEPS

```
Source
Removed top rows
Promoted headers
Filtered out blank rows
Removed duplicates
Split column by delimiter
Added conditional column
Changed type
Merged queries
```

1,535 rows in, **1,500 clean rows** out

File: raw_client_export_Q4_2025.csv (1,500 orders, $314,348 of Q4 2025 sales once cleaned)

Every click is a line of code you would have written

Click in Power Query	M code it records	dplyr twin
Remove top rows	Table.Skip	read_csv(skip=3)
Remove duplicates	Table.Distinct	distinct()
Split column	Table.SplitColumn	separate()
Conditional column	Table.AddColumn	case_when()
Change type	Table.TransformColumnTypes	as.Date()
Merge queries	Table.NestedJoin	left_join()

The steps replay on every refresh. Next month's file gets the same cleaning with nobody touching it.

A measure is written once and answers any filter

```
Revenue =
  SUM ( Sales[Revenue] )
```

Think of it as a **reactive summarise()** that reruns for every card, bar and cell, using whatever filters touch it.

No filter
$14.39M

Region = Southwest
$2.89M

Year = 2026
$4.82M

Southwest · Camping · Aug 2026
$14.2K

Measure = summarise(): computed on the fly, never stored

Calculated column = mutate(): row by row, stored in the table

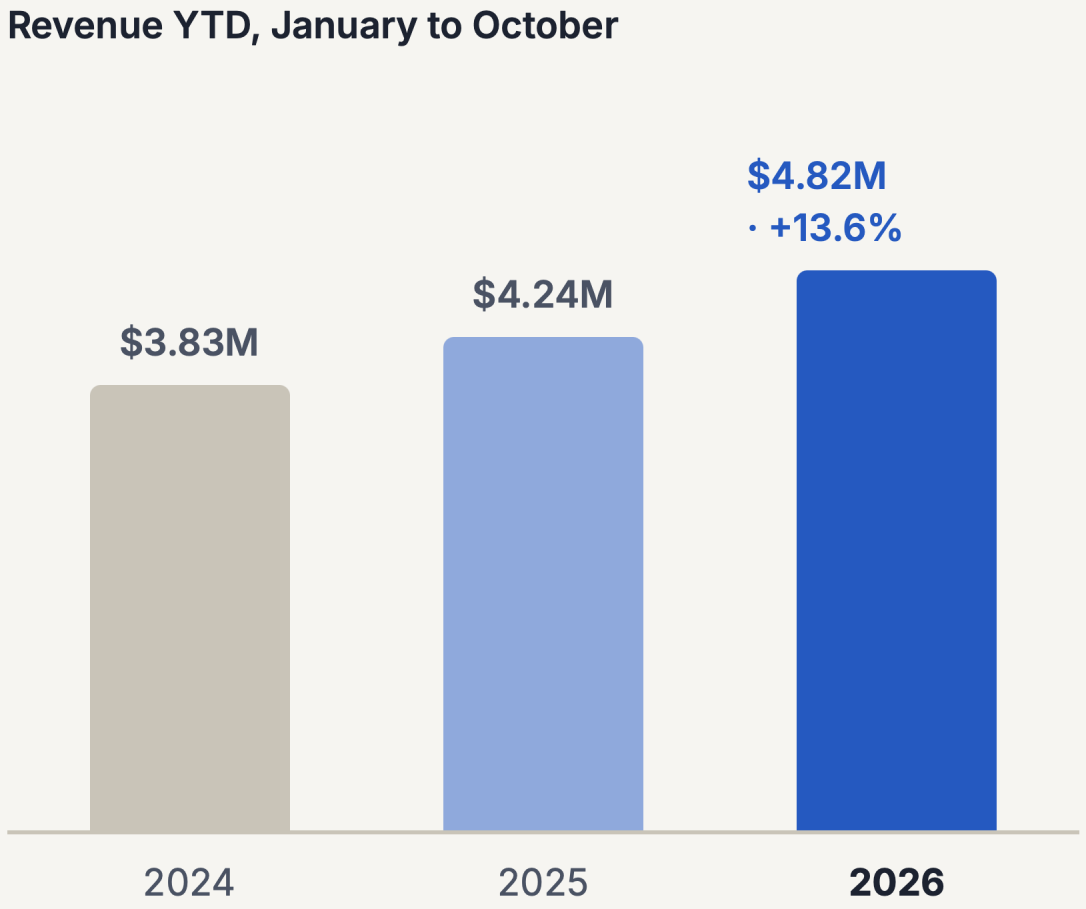
CALCULATE and time intelligence

```
Revenue LY =
  CALCULATE ( [Revenue],
    SAMEPERIODLASTYEAR (
      'Calendar'[Date] ) )

YoY % =
  DIVIDE ( [Revenue] - [Revenue LY],
    [Revenue LY] )

Revenue YTD =
  TOTALYTD ( [Revenue], 'Calendar'[Date] )
```

CALCULATE = run a measure with different filters. It powers almost every advanced DAX pattern.



Needs a Calendar table marked as the date table. All measures are in dax_measures.dax.

05

Dashboards with drill-down

From “something is off” to the exact cause in four clicks

Revenue peaks every summer and grows each year



Summer peaks

Camping and water gear drive June to August

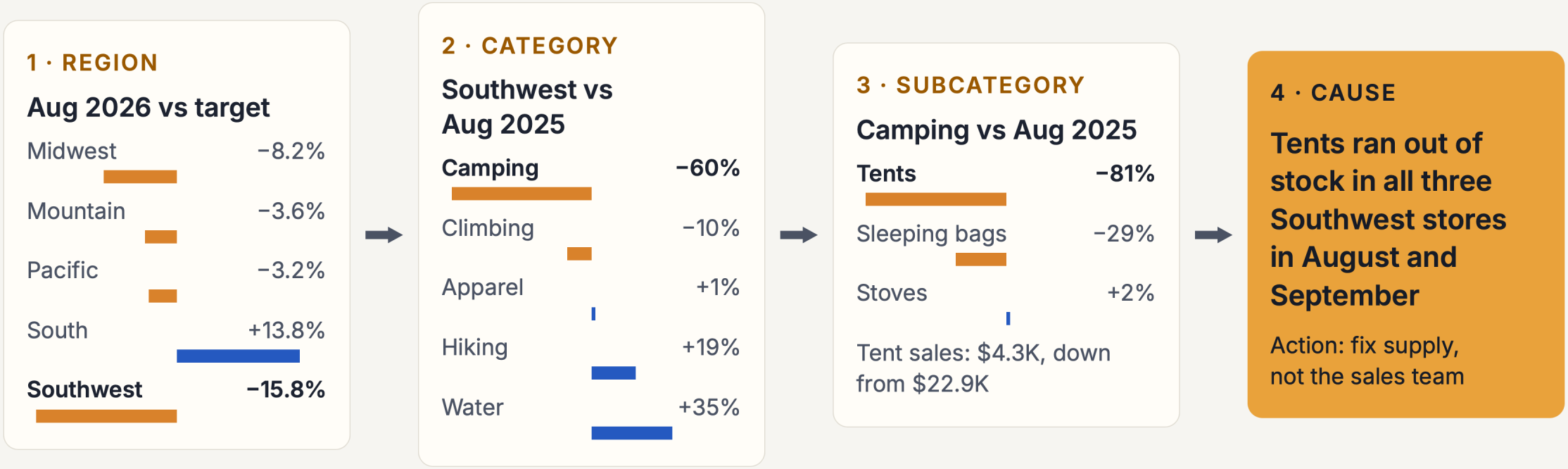
December bump

Jackets and gifts lift the holiday month

Growing

Each summer’s peak beats the last

Four clicks: why did the Southwest miss August?



Orange bars are below target or last year; blue bars are above. Every label carries its sign.

Four ways to explore, none of them coded

Drill down

Stack fields on one axis and use the arrows: Year → Quarter → Month, Region → State → Store

Cross-filter

Click any bar and every other visual on the page filters to it. Edit interactions to change that.

Drill through

Right-click a store and jump to a detail page already filtered to that store

Report tooltips

Hover over a point to see a small report page, such as a store's monthly trend

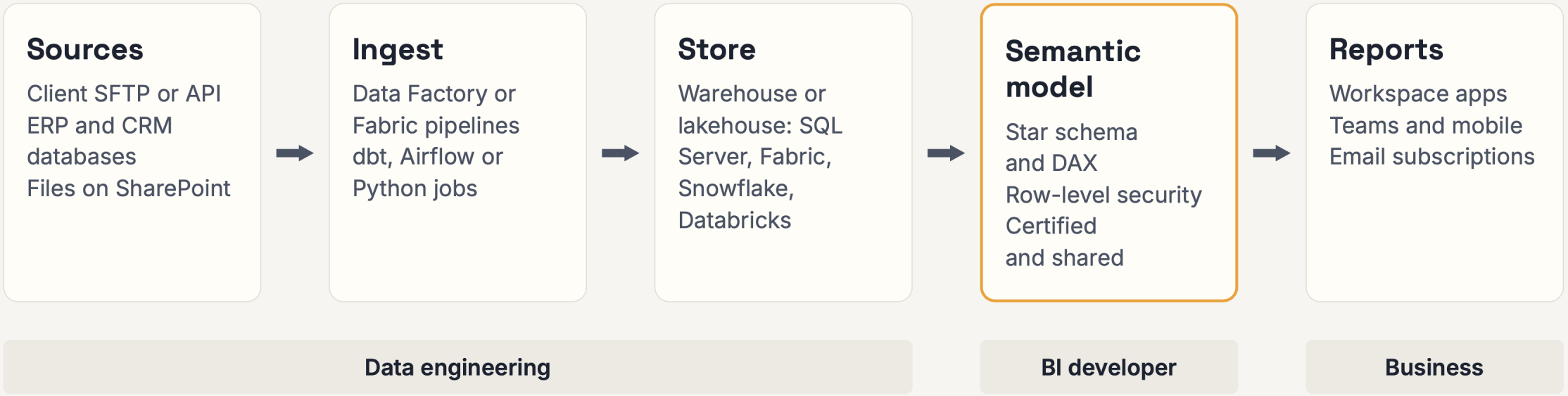
In Shiny, each of these is an event handler, a reactive value and a back button.

06

How large companies run it

From the client’s file drop to a refreshed report before breakfast

The enterprise pipeline, end to end



Data behind the firewall? An on-premises data gateway, a Windows service inside the network, lets cloud refreshes reach it over an outbound-only connection.

One night in the life of an automated refresh

01:00

Client drops a file

sales_2026_11.csv lands on SFTP or in a SharePoint folder

01:30

Pipeline loads it

Into the warehouse, with checks on row counts and duplicates

02:00

Model refreshes

On a schedule, or triggered by the pipeline when the load succeeds

02:05

Only new days reload

Incremental refresh leaves years of history untouched

07:00

People see it

Subscriptions email the report; alerts fire if a KPI drops

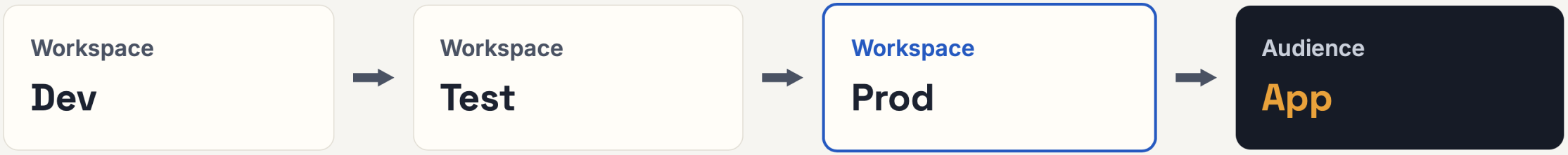
⚠ **When it breaks:** the owner gets a failure email, and after four failures in a row Power BI switches the schedule off until someone fixes it.

Four ways to keep data fresh

Option	How it works	Freshness	Watch out for
Scheduled refresh	The Service reloads the model at times you pick	8 a day on Pro, 48 on PPU or Fabric	Gateway for on-prem data
Triggered refresh	The pipeline calls the refresh API when its load finishes	Minutes after new data lands	Counts toward the same daily limit
Incremental refresh	Table split by date; only recent partitions reload	Same schedule, much faster	Needs a date column and two parameters
DirectQuery or Direct Lake	No import; each visual queries the source live	Near real time	Source speed, some DAX limits

Cloud files need no gateway. Files on OneDrive or SharePoint refresh straight from the Service, which is how today's Mac demo works.

Dev, test, prod, and who sees what



Deployment pipelines promote content stage by stage, swapping test data sources for production ones.

Row-level security

One report; each regional manager sees only their own region

Certified models

Endorsed semantic models give everyone one definition of revenue

Git and PBIP

Reports saved as text files and reviewed in pull requests, like code

When a new client file appears

1. Put the Aug, Sep and Oct files from **client_drops** in a OneDrive or SharePoint folder
2. Create › Get data › **SharePoint folder** › filter to the folder › Combine
3. Line chart of revenue by day: it stops at October 31
4. Semantic model settings › sign in to the source › **Scheduled refresh** at 07:00
5. Drop **NEW_sales_2026_11.csv** into the folder, as the client would
6. **Refresh now**: November 1–10 appears, with no change to the report

BEFORE

6,791 orders

Aug – Oct 2026 · $1.60M

AFTER REFRESH

7,462 orders

+671 orders · +$130K from Nov 1–10

Cloud folder, so no gateway. In a company the folder is the client's SFTP drop or a warehouse table, and step 6 is the nightly schedule.

Six things to remember

1. Use a work or school email. **Free builds, Pro shares.**
2. Shiny for custom apps, **Power BI for reporting at scale**
3. Build the **semantic model** once; many reports can share it
4. Power Query **records** the cleaning; DAX measures **react** to filters
5. Drill-down lets people answer **“why?”** on their own
6. Companies chain **pipeline** → **warehouse** → **refresh**, so nobody clicks refresh

Questions?

The data, the Shiny and Streamlit apps, the DAX and the M scripts are all in the talk pack.

Keep learning

Microsoft Learn
free Power BI learning paths

dax.guide
every DAX function with examples

SQLBI
the deep dives on modelling and DAX

Power Query M reference
when the clicks are not enough